Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Correctly identifies the complex conjugate as a root.	AO1.1b	B1	2nd complex root is $2 + 3\mathrm{i}$ sum of roots $= -\frac{b}{a}$ $\therefore \alpha + 2 + 3\mathrm{i} + 2 - 3\mathrm{i} = 0$ $\alpha + 4 = 0$ $\alpha = -4$
	Forms one of the following equations (or their equivalents) $\alpha + \beta + 2 - 3\mathrm{i} = 0$ or $\alpha\beta(2 - 3\mathrm{i}) = \pm 52$ or $\alpha\beta + \beta(2 - 3\mathrm{i}) + (2 - 3\mathrm{i})\beta = \pm m$ with an equation to find m. Or forms an equation to find m and then solves the cubic for their value of m.	AO1.1a	M1	
	Finds correct third root	AO1.1b	A1	
				2nd complex root is $2 + 3\mathrm{i}$ product of roots $= -\frac{d}{a}$ $\therefore \alpha(2 + 3\mathrm{i})(2 - 3\mathrm{i}) = -52$ $\alpha(4 - 9\mathrm{i}^2) = -52$ $\alpha = \frac{-52}{13}$ $\alpha = -4$
(b)	Forms a correct equation to find m. May be seen in part (a).	AO1.1a	M1	$\therefore (-4)^3 + m(-4) + 52 = 0$
	Finds correct value of m.	AO1.1b	A1	$-64 - 4m + 52 = 0$ $-12 = 4m$ $m = -3$
				$\sum \alpha\beta = \frac{c}{a}$ $\therefore (2 - 3\mathrm{i})(2 + 3\mathrm{i}) + (-4)(2 - 3\mathrm{i}) + (-4)(2 + 3\mathrm{i}) = m$ $4 + 9 - 8 + 12\mathrm{i} - 8 - 12\mathrm{i} = m$ $m = -3$
				Total 5 marks

Q2.

Marking Instructions	AO	Marks	Typical Solution

(a)	Makes a correct deduction about another root (PI)	AO2.2a	B1	$(z-(2-3\mathrm{i}))(z-(2+3\mathrm{i})) = z^2-4z+13$
	Finds quadratic factor by expanding brackets or using sum and product of roots	AO1.1a	M1	$\therefore$ $p(z) = (z^2-4z+13)(z^2+cz+d)$ $(z^2-4z+13)(z^2+cz+d) \equiv z^4+3z^2+az+b$
	Finds a correct quadratic factor	AO1.1b	A1	
	Compares coefficients with quartic z^4+3z^2+az+b	AO1.1a	M1	$c-4=0$ $13-4c+d=3$ $\Rightarrow c=4,\ d=6$
	States the correct product of quadratic factors	AO1.1b	A1	$\therefore$ $p(z) = (z^2-4z+13)(z^2+4z+6)$
ALT	Makes a correct deduction about another root	AO2.2a	B1	$(z-(2-3\mathrm{i}))(z-(2+3\mathrm{i})) = z^2-4z+13$ $\therefore$
	Finds quadratic factor by expanding brackets or using sum and product of roots	AO1.1a	M1	$p(z) = (z^2-4z+13)(z^2+cz+d)$ $(z^2-4z+13)(z^2+cz+d) \equiv z^4+3z^2+az+b$
	Obtains a correct quadratic factor	AO1.1b	A1	$\alpha+\beta+\gamma+\delta=0 \Rightarrow \gamma+\delta=-4$ $\therefore c=4$
	Uses coefficients/roots to set up equations and find required coefficients	AO1.1a	M1	$(\sum\alpha)^2 = \sum\alpha^2 + 2\sum\alpha\beta$ $0 = \sum\alpha^2 + 2\times 3$ $0 = (2-3\mathrm{i})^2+(2+3\mathrm{i})^2+\gamma^2+\delta^2+6$
	States the correct product of quadratic factors	AO1.1b	A1	$\therefore \gamma^2+\delta^2=4$ $2\gamma\delta = (\gamma+\delta)^2-\gamma^2+\delta^2$ $\gamma\delta = \frac{16-4}{2}$ $\therefore d=6$ $p(z) = (z^2-4z+13)(z^2+4z$

				+ 6)

(b)	States all four correct solutions FT 'their' two quadratic factors from part (a) provided both M1 marks have been awarded	AO1.1b	B1F	$z = 2 \pm 3\mathrm{i}, -2 \pm \sqrt{2}\mathrm{i}$
				Total 6 marks

Q3.

(a) $(w=)\ \dfrac{-6 \pm \sqrt{36 - 4(34)}}{2}$ $\left\{= \dfrac{-6 \pm \sqrt{-100}}{2}\right\}$

Correct substitution into quadratic formula OE

M1

$= \dfrac{-6 \pm 10\mathrm{i}}{2}$

$\sqrt{-100}$ *= 10i or* $\sqrt{-100}$ */ 2 = 5i*

B1

= −3 ± 5i

−3 ± 5i (p= −3, q = ±5)

NMS mark as 3/3 or 0/3

A1

3

(b) (i) $z = \mathrm{i}(1 + \mathrm{i})(2 + \mathrm{i}) = \mathrm{i}(2 + 3\mathrm{i} + \mathrm{i}^2) = 2\mathrm{i} + 3\mathrm{i}^2 + \mathrm{i}^3$

Attempt to expand all brackets.

M1

= 2i + 3(−1) + i(−1)

$i^2 = -1$ used at least once

B1

= −3 + i

−3 + i (a = −3, b = 1)

A1

3

(ii) $z^* = -3 - \mathrm{i}$

$-3 + \mathrm{i} + m(-3 - \mathrm{i}) = n\mathrm{i}$

OE Ft c's a − bi

B1F

$\Rightarrow -3 - 3m = 0$; $1 - m = n$

*Equating **both** real parts and the imag. parts, PI by next line*

M1

$\Rightarrow m = -1$, $n = 2$

Both correct

A1

3

[9]

Q4.

(a) (i) $(z - 2i)^* = (x + yi - 2i)^* = x + (2 - y)i$

$x + 2i - yi$ OE rearrangement

B1

1

(ii) $(z - 2i)^* = 4iz + 3 = 4ix + 4i^2y + 3 = 4ix - 4y + 3$

$x + (2 - y)i = 4ix - 4y + 3$ (#)

$i^2 = -1$ used

B1

Real parts: $x = -4y + 3$

Imaginary parts: $2 - y = 4x$

*Attempting to equate, without mixing real and imaginary terms, **both** the real parts and the imag. parts for the c's eqn (#).*

M1

If not corrected, ft on [c's(a)(i)] = 4ix − 4y + 3 provided both the resulting linear equations have non zero x, y and const terms

A1F

$y = \frac{2}{3}$, $x = \frac{1}{3}$

Solving correct equations, to obtain either $x = \frac{1}{3}$ OE or $y = \frac{2}{3}$ OE

A1

$(z =)\ \frac{1}{3} + \frac{2}{3}i$

$\frac{1}{3} + \frac{2}{3}\,i$

A1 **5**

(b) (One of the) coefficients (of the quadratic equation is) not real.

OE eg Sum of roots is −10 i so p cannot be real if roots are p ± qi

E1 **1**

[7]

Q5.

(a) (i) $x = \pm\, 3\text{i}$

± 3i (a = 0, b = ±3)

B1 **1**

(ii) $x = -2 \pm 3\text{i}$

If not correct, ft wrong answer(s) to (i) provided (i) has a non-zero b value

B1F **1**

(b) (i) $(1 + x)^3 = 1 + 3x + 3x^2 + x^3$

Terms simplified in any order.

B1

(ii) $(1 + 2\text{i})^3 = 1 + 3(2\text{i}) + 3(2\text{i})^2 + (2\text{i})^3 = 1 + 3(2\text{i}) + 3(4\text{i}^2) + (8\text{i}^3)$

Replacing x in (b)(i) by 2i, squaring and cubing correctly, only ft on c's wrong non-zero coefficients from (b)(i).

B1F

$= 1 + 3(2\text{i}) + 3(4)(-1) + (8)(-\text{i})$

Use of $i^2 = -1$ at least once.

M1

$= -11 - 2\text{i}$

−11 − 2i (a = −11, b = −2)

A1 **3**

(iii) $z^* - z^3 = 1 - 2\text{i} - (-11 - 2\text{i}\,)$

Use of $z^* = 1 - 2i$

M1

= 12

If not correct, only ft on 1 − 2i − c's (b)(ii) if c's (b)(ii) answer is of the form a + bi with $a \neq 0$ and $b \neq 0$

A1F

2

[8]

Q6.

(a) (i) $4(1+\mathrm{i}\sqrt{3}) = 8\left(\frac{1}{2}+\mathrm{i}\frac{\sqrt{3}}{2}\right)$

for either $4(1+\mathrm{i}\sqrt{3})$ *or* $4(1-\mathrm{i}\sqrt{3})$ *used*

M1

$= 8e^{\frac{\pi i}{3}}$

If either r or θ is incorrect but the same value in both (i) and (ii) allow A1 but for θ only if it is given as $\frac{\pi}{6}$

A1

(ii) $4(1-\mathrm{i}\sqrt{3}) = 8e^{\frac{-\pi i}{3}}$

A1

3

(b) $z^3 - 4 = \pm\sqrt{-48}$

taking square root

M1

$z^3 = 4 \pm 4\sqrt{3}\mathrm{i}$

AG

A1

2

(c) (i) $z = 2e^{\frac{\frac{\pi i}{3}+2k\pi i}{3}}$ or $z = 2e^{\frac{\frac{-\pi i}{3}+2k\pi i}{3}}$

for the 2; ft incorrect 8, but no decimals

B1F

for either, PI

M1

$z = 2e^{\frac{\pi i}{9}}, 2e^{\frac{7\pi i}{9}}, 2e^{\frac{5\pi i}{9}}$

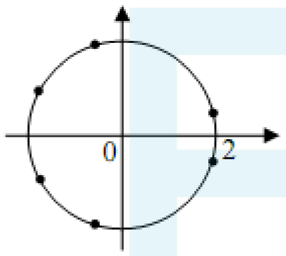
$2e^{\frac{-\pi i}{9}}, 2e^{\frac{-7\pi i}{9}}, 2e^{\frac{-5\pi i}{9}}$

Allow A1 for any 2 roots not +/– each other
Allow A2 for any 3 roots not +/– each other
Allow A3 for all 6 correct roots
Deduct A1 for each incorrect root in the interval; ignore roots outside the interval
ft incorrect r

A3,2,1F

5

(ii)



Radius 2

clearly indicated; ft incorrect r
allow B1 for 3 correct points condone lines

B1F

Plotting roots

B2,1

3

(d) (i) Sum of roots = 0 as coefficient of $z^5 = 0$

OE

E1

1

(ii) Use of, say, $\frac{1}{2}\left(e^{\frac{\pi i}{9}} + e^{\frac{-\pi i}{9}}\right) = \cos\frac{\pi}{9}$

M1

$\cos\frac{3\pi}{9} = \frac{1}{2}$ used

A1

$\cos\frac{\pi}{9} + \cos\frac{3\pi}{9} + \cos\frac{5\pi}{9} + \cos\frac{7\pi}{9} = \frac{1}{2}$

AG

A1 3

[17]

Q7.

(a) (i) $\alpha + \beta = 2$, $\alpha\beta = 4$

B1B1

$\alpha^3 + \beta^3 = (2)^3 - 3(4)(2) = -16$

M1A1

$\alpha^3 \beta^3 = (4)^3 = 64$, hence result

convincingly shown (AG)

M1A1 6

(ii) Discriminant 0, so roots equal

or by factorisation

B1E1 2

(b) $x = \dfrac{2 \pm \sqrt{4-16}}{2}$

or by completing square

M1

$\ldots = 1 \pm \frac{1}{2}\mathrm{i}\sqrt{12}$

A1 2

(c) $\alpha, \beta = 1 \pm \mathrm{i}\sqrt{3}$

and $\alpha^3 = \beta^3$, hence result

E2 2

[12]

Q8.

(a) (i) Roots are $\pm 4\mathrm{i}$

M1 for one correct root or two correct factors

M1A1 2

(ii) Roots are $1 \pm 4i$

M1 for correct method

M1A1 2

(b) (i) $(1+x)^3 = 1 + 3x + 3x^2 + x^3$

M1A0 if one small error

M1A1 2

(ii) $(1+i)^3 = 1 + 3i - 3 - i = -2 + 2i$

M1 if $i^2 = -1$ used

M1A1 2

(iii) $(1+i)^3 + 2(1+i) - 4i$

with attempt to evaluate

M1

$\ldots = (-2 + 2i) + (2 - 2i) = 0$

*convincingly shown (**AG**)*

A1 2

[10]

Q9.

(a) $\sum \alpha\beta = 6$

B1 1

(b) (i) Sum of squares < 0 $\therefore$ not all real

E1

Coefficients real $\therefore$ conjugate pair

E1 2

(ii) $\left(\sum \alpha\right)^2 = \sum \alpha^2 + 2\sum \alpha\beta$

A1 for numerical values inserted

M1A1

$\left(\sum \alpha\right)^2 = 0$

A1F

$p = 0$

cao

A1F

4

(c) (i) $-1 - 3i$ is a root

B1

Use of appropriate relationship eg $\sum \alpha = 0$

M0 if $\sum \alpha^2$ *used unless the root 2 is checked*

M1

Third root 2

incorrect $p\sqrt{}$

A1F

3

(ii) $q = -(-1 - 3i)(-1 + 3i)^2$

allow even if sign error

M1

$= -20$

ft incorrect 3rd root

A1F

2

[12]

Q10.

(a) (i) $z^3 = \dfrac{4 \pm \sqrt{16 - 32}}{2}$

M1

$= 2 \pm 2i$

AG

A1

2

(ii) $2 + 2i = 2\sqrt{2}e^{\frac{\pi i}{4}}$, $2 - 2i = 2\sqrt{2}e^{\frac{-\pi i}{4}}$

M1 for either result or for one of $r = 2\sqrt{2}$, $\theta = \pm\dfrac{\pi}{4}$

M1

$$\left(r = 2\sqrt{2} \quad A1,\ \theta = \pm\frac{\pi}{4}\, A1\right)$$

A1A1

$$z = \sqrt{2}e^{\frac{\pi i}{12}+\frac{2k\pi i}{3}} \text{ or } \sqrt{2}e^{\frac{-\pi i}{12}+\frac{2k\pi i}{3}}$$

M1 for either

M1

$$z = \sqrt{2}e^{\frac{\pm\pi i}{12}}, \sqrt{2}e^{\frac{\pm3\pi i}{4}}, \sqrt{2}e^{\frac{\pm7\pi i}{12}}$$

allow A1 for any 3 correct ft errors in $\pm\frac{\pi}{4}$

A2,1,0 F

6

(b) Multiplication of brackets

M1

Use of $e^{i\theta} + e^{-i\theta} = 2\cos\theta$

AG

A1

2

(c) $$\left(z - \sqrt{2}e^{\frac{\pi i}{12}}\right)\left(z - \sqrt{2}e^{-\frac{\pi i}{12}}\right)$$

$$= z^2 - 2\sqrt{2}\cos\tfrac{\pi}{12}z + 2$$

PI

M1A1F

$$\left(z^2 - 2\sqrt{2}\cos\tfrac{\pi}{12}z + 2\right)$$

Product is $\left(z^2 - 2\sqrt{2}\cos\tfrac{7\pi}{12}z + 2\right)$

(or $z^2 + 2z + 2$)

A1F

3

$(z^2 - 2\sqrt{2}\cos\frac{3\pi}{4}z + 2)$

[13]

Q11.

(a) $p = -4$

B1

$(\alpha + \beta + \gamma)^2 = \Sigma\alpha^2 + 2\Sigma\alpha\beta$

M1

$16 = 20 + 2\Sigma\alpha\beta$

A1

$\Sigma\alpha\beta = -2$

A1F

$q = -2$

A1F

5

(b) 3 − i is a root

B1

Third root is −2

B1F

$\alpha\beta\gamma = (3 + i)(3 - i)(-2)$

M1

$= -20$

Real $\alpha\beta\gamma$

A1F

$r = +20$

Real r

A1F

5

Alternative to (b)
Substitute 3 + i into equation

M1

$(3 + i)^2 = 8 + 6i$

B1

$(3 + i)^3 = 18 + 26i$

B1

$r = 20$

Provided r is real

A2,1,0

[10]

Q12.

(a) (i) $z + \frac{1}{z} = \cos\theta + \mathrm{i}\sin\theta + \cos(-\theta) + \mathrm{i}\sin(-\theta)$

Or $z + \frac{1}{z} = e^{i\theta} + e^{-i\theta}$

M1

$= 2\cos\theta$

AG

A1

2

(ii) $z^2 + \frac{1}{z^2} = \cos 2\theta + i\sin 2\theta + \cos(-2\theta) + i\sin(-2\theta)$

M1

$= 2\cos 2\theta$

OE

A1

2

(iii) $z^2 - z + 2 - \frac{1}{z} + \frac{1}{z^2}$

$= 2\cos 2\theta - 2\cos\theta + 2$

M1

Use of $\cos 2\theta = 2\cos^2\theta - 1$

m1

$= 4\cos^2\theta - 2\cos\theta$

AG

A1

3

(b) $z + \frac{1}{z} = 0$ $z = \pm\mathrm{i}$

M1A1

$z + \frac{1}{z} = 1 \qquad z^2 - z + 1 = 0$

M1A1

$z + \frac{1 \pm i\sqrt{3}}{2}$

A1F

Alternative:

$\cos\theta = 0 \qquad \theta = \pm \frac{1}{2}\pi$

M1

$z = \pm i$

A1

$\cos\theta = \frac{1}{2} \qquad \theta = \pm \frac{1}{3}\pi$

M1

$z = e^{\pm\frac{1}{3}\pi i} = \frac{1}{2}(1 \pm i\sqrt{3}$

A1A1

5

Accept solution to (b) if done otherwise

Alternative

If $\theta = + \frac{1}{2}\pi \quad \theta = \frac{1}{3}\pi$

M1

$z = i \quad z = \frac{1 + \sqrt{3}i}{2}$

A1

Or any correct z values of θ

M1

Any 2 correct answers

A1

One correct answer only

B1

[12]

Q13.

(a) Asymptotes $x = 0,\ y = 1$

B1, B1

2

(b) (i) $\Delta = 4 - 8 < 0$, so num never 0

OE; E1 for incomplete explanation

E2,1

2

(ii) Method for solving quadratic

"i" must appear

M1

Roots –1 ± i

A1 if one error made

A2,1

3

(c) (i) $f(x) = k \Rightarrow x^2 + 2x + 2 = kx^2$

M1

$\ldots \Rightarrow (1 - k)x^2 + 2x + 2 = 0$

m1

Equal roots $\Rightarrow 4 - 8(1 - k) = 0$

Convincingly shown (AG)

A1

3

(ii) $k = \frac{1}{2}$

B1

$y = \frac{1}{2}$ ***at SP***

ft wrong value for k

B1ft

So $\frac{1}{2}x^2 + 2x + 2 = 0$

M1

and $x = -2$ at SP

A1

4

[14]

Examiner reports

Q1.

This question was well answered with almost every student identifying the complex conjugate as a root of the equation. The methods for finding the third root were many and varied, as were the methods for finding the value of m. However, the majority of attempts were successful. Some students opted to find the three factors, expand the brackets and then compare coefficients. Although this is a perfectly valid method, it was quite time consuming, and prone to error.

A few students found m whilst calculating the third root. Many of these then wasted time in rewriting their method in part (b). Some of the less successful attempts confused factors with roots.

Q3.

Many correct answers were submitted for parts (a) and (b)(i) in this question which tested the complex numbers section of the specification. In part (b)(ii) a common mistake was anincorrect z^* value either $i + 3$ following $z = i - 3$ in (b)(i) or, more frequently, z^* written incorrectl as $i(1 - i)(2 - i)$. Students who obtained an incorrect value for z in part (b)(i) were awardedcredit for a correct follow through value for z^* in part (b)(ii). Most students showed that theyappreciated the need to equate the real parts and the imaginary parts for their resulting equation.

Q4.

This question which tested the complex numbers section of the specification proved, as expected, to be more demanding than the corresponding questions in recent MFP1 papers. In part (a)(i) a common wrong answer for ($z - 2\text{i}$)* was $x + \text{i}\,y + 2\text{i}$. Those students who started by writing $(x + \text{i}y - 2\text{i})^*$ were generally more successful. There were some very good solutions to part (a)(ii), but some able students, having found the correct values for x and y, either stopped at that point or else gave their final answer as $\frac{1}{3}+\frac{4}{3}\text{i}$ which indicated that they thought that solving the given equation meant finding the value of $(z - 2\text{i})^*$ and not the value of z. It was quite common to see some students making no attempt to equate real parts and imaginary parts, with final answers left in a form similar to $x + 4y - 3 + (2 - y - 4x)\text{i}$. The final part of the question was answered well by those who based their statement on:

- coefficients must be real for conjugate roots or
- sum of roots ($= 2p$) cannot be -10i if p is real or
- product of the roots ($p^2 + q^2$) cannot be -29 if p and q are real.

However, correct statements were in the minority. Common wrong answers for example, 'it is not the difference of two squares' suggested that students were treating the roots $p \pm \text{i}q$ as the factors of the quadratic expression $z^2 + 10\text{i}z - 29$.

Q5.

In this question on complex numbers, part (a) generally caused more problems than part (b).

In part (a)(i) a significant number of candidates failed to include the ±, but, as follow through was accepted, most of these scored the mark in part (a)(ii).

Almost all candidates correctly expanded $(1 + x)^3$ in part (b)(i) and used their expansion within part (b)(ii) but there were many examples of incorrect / no squaring and cubing of the 2. The meaning and use of the complex conjugate z^* was well understood and a majority of candidates went on to give a correct follow through answer in part (b)(iii).

Q6.

Parts (a) and (b) of this question were well done; part (c)(i) was reasonably well done especially by the more able candidates. However, the Argand diagrams were very poorly drawn. Many candidates drew a rough circle by hand, with no indication of its radius, and then went on to put crosses on their circle at points which bore little resemblance to the angles in question. Some candidates drew no circle at all but drew lines of differing lengths from the origin. There were few correct responses to part (d) as candidates failed to realise the relevance of part (c), and especially part (d)(i), to this last part.

Q7.

This question proved to be an excellent source of marks for the majority of candidates, apart from the discriminating test provided by part (c). Many candidates seemed to be familiar with the techniques relating to the cubes of the roots of a quadratic equation, though some struggled to find the correct expression for the sum of the cubes of the roots, while others lost marks by not showing enough evidence in view of the fact that the required equation was printed on the question paper.

Parts (a)(ii) and (b) proved straightforward for most candidates, but only a few were able to give a clear explanation in part (c), where many candidates claimed that the original roots α and β must be equal.

Q8.

This opening question gave almost all the candidates the opportunity to score a high number of marks. Even when careless errors were made, for example the omission of the plus-or-minus symbol in one or both sections of part (a), there was much correct work for the examiners to reward. The expansion of the cube of a binomial expression in part (b)(i) seemed to be tackled more confidently than in the past. Almost all the candidates used $i^2 = -1$ in part (b)(ii), though i^3 some were unsure how to deal with.

Q9.

Whilst part (a) was usually correctly done, part (b)(i) was poorly answered. Some candidates were able to comment on the condition that as the sum of the squares of the roots was less than zero there would have to be complex roots, but few stated the conditions that the coefficients of the cubic equation were all real. The value of p in part (b)(ii) was very often correct but in part (c)(i) a very common error as to use $\sum \alpha^2 = -12$ in order to find the third root. This method led to $\alpha^2 = 4$ from which almost all candidates using this method wrote $\alpha = 2$ without even considering the possibility that α could equal –2. Part (c)(ii) was usually worked correctly although $\alpha\beta\gamma = +q$ appeared from time to time.

Q10.

Candidates were usually able to establish the result in part (a) although the methods used were sometimes somewhat inelegant. Part (a)(ii) was reasonably well done although some carelessness was in evidence in this part. For instance, some candidates although showing that the argument of z^3 was $\pm\frac{1}{4}\pi$ continued their solution with only $+\frac{1}{4}\pi$ and so arrived at a total of three roots. Others having reached $|z^3| = \sqrt{8}$ then thought that $|z| = \sqrt{8}$ also.

A few candidates used a method which, although possible, was not really suitable. They replaced the z^3 in $z^6 - 4z^3 + 8 = 0$ with $2 \pm 2i$ and so arrived at $z^6 = \pm 8i$. This latter equation gave the twelve roots of $z^{12} = -64$ and the method was incomplete unless 6 of the roots were rejected. Part (b) was generally well done, but part (c) was really only completed by candidates who had correctly answered part (a)(ii).

Q11.

Candidates were also able to achieve good results on this question. In part (a), where errors occurred they were almost always errors of signs. For instance the value of p was given as 2 instead of –2 or the formula for $(\Sigma\,\alpha)^2$ was incorrectly quoted as $\Sigma\alpha^2 - 2\Sigma\,\alpha\beta$. There were fewer completely correct solutions to part (b) often due to inelegant methods of solution. The most successful candidates obtained the values of the other two roots and then worked out the product $\alpha\beta\gamma$. The main loss of marks using this method was to equate r to $\alpha\beta\gamma$ instead of $-\alpha\beta\gamma$. The other main method of approach to this part of the question was to substitute 3 + i into the cubic equation with the values of p and q already found in part (a). However any error in the values of p and q or in the substitution inevitably led to r having a complex value. Surprisingly this did not seem to worry the candidates in spite of the fact that the question stated that r was real.

Q12.

Parts (a) was quite well done and many candidates scored the available seven marks. There were however some serious algebraic errors, the commonest of which was to equate $z^2 + \frac{1}{z^2}$ to $\left(z + \frac{1}{z}\right)^2$ in part (a)(ii) with some consequent faking to arrive at the printed result in part (a)(iii). Part (b) however was very poorly attempted. Candidates did not seem to realise that z could be equal to zero and consequently multiplied $(\cos\theta + i\sin\theta)^2$ by $4\cos^2\theta - 2\cos\theta$ Of those candidates who realised that $4\cos^2\theta - 2\cos\theta$ was equal to zero, the factorisation of this quadratic in $\cos\theta$ evaded most and, even when attempts were made to solve $4\cos^2\theta - 2\cos\theta = 0$, the factor $\cos\theta$ disappeared and the other solution $\cos\theta = 1/2$ usually produced just one root from $\theta = \frac{\pi}{3}$. Candidates who were able to obtain both $\cos\theta = 0$ and $\cos\theta = 1/2$ usually produced only two solutions and subsequently two roots. It did not seem to occur to candidates that a quartic equation would have four roots.

Q13.

Part (a) was generally well answered, although the equation $y = 0$ was often given instead of $y = 1$ for the horizontal asymptote. Attempts at part (b)(i) were often inconclusive, but part (b)(ii) was often well attempted despite errors in simplifying the expression $\frac{-2 \pm \sqrt{-4}}{2}$. Part (c) was only seriously attempted by stronger candidates, of whom many produced concise accurate solutions in both parts. No credit was given for attempts to use differentiation to find the stationary point.